

Frontend – Assignment 1

Q1. HTTP Request–Response Cycle

The HTTP Request–Response Cycle is the process through which a browser communicates with a web server.

Diagram:

Client/Browser → HTTP Request → Web Server → HTTP Response → Client/Browser

Steps:

1. The user enters a URL or clicks a link.
2. The browser sends an HTTP request to the server.
3. The server processes the request.
4. The server sends an HTTP response containing the requested resource.
5. The browser displays the received content.

Q2. Introduction to Web Technologies

1. HTML – Creates the structure and content of a web page.
2. CSS – Adds styling, layout, colors, fonts, and responsiveness.
3. JavaScript – Adds logic, interaction, and dynamic behavior.
4. Bootstrap – Provides ready-made responsive UI components and CSS classes.
5. jQuery – A JavaScript library that simplifies DOM manipulation and events.
6. React.js – A JavaScript library for building reusable and interactive UI components.

Q3. Client-Side vs. Server-Side Technologies

Client-side technologies run mainly in the user's browser. Examples: HTML, CSS, JavaScript.

Server-side technologies run on the web server and handle data, logic, and requests. Examples: Java, Python, PHP, Node.js.

Main difference:

Client-side → Browser → User interface and interaction.

Server-side → Server → Business logic, database, and request processing.

Q4. Block-Level Elements in HTML

Block-level elements start on a new line and normally take the full available width of their container. They can contain other elements and are useful for structuring a page.

Examples: <div>, <p>, <h1>, <section>.

Q5. Semantic HTML Elements

Semantic elements clearly describe the meaning and purpose of their content. They make HTML easier to understand and improve accessibility and SEO.

Examples: <header>, <nav>, <main>, <section>, <article>, <footer>.

Q6. Creating a Table Using HTML

HTML code:

```
<table border="1">
  <tr>
    <th>Sr. No.</th><th>Name</th><th>Designation</th><th>Department</th>
  </tr>
  <tr><td>1</td><td>John</td><td>Manager</td><td>Sales</td></tr>
  <tr><td>2</td><td>Smith</td><td>Sr. Manager</td><td>Marketing</td></tr>
  <tr><td>3</td><td>Jane</td><td>Manager</td><td>HR</td></tr>
</table>
```

Q7. Creating a Nested List Using HTML

HTML code:

```
<ol>
  <li>River
    <ul>
      <li>Ganga</li>
      <li>Yamuna</li>
      <li>Brahmaputra</li>
      <li>Narmada</li>
    </ul>
  </li>
```


Q8. Types of CSS

1. Inline CSS – Written directly inside an HTML element.

Example: <p style="color: blue;">Hello</p>

2. Internal CSS – Written inside a <style> tag in the HTML document.

Example:

<style>

p { color: blue; }

</style>

3. External CSS – Written in a separate .css file and linked to HTML.

Example:

<link rel="stylesheet" href="style.css">

Q9. Practical Understanding

HTML – Provides the structure of the webpage, such as headings, paragraphs, images, and forms.

CSS – Controls the appearance and layout, such as colors, fonts, spacing, and responsiveness.

JavaScript – Adds behavior and interactivity, such as buttons, validation, and dynamic content.

Q10(a). What is the role of Bootstrap in web development?

Bootstrap is a frontend framework that provides ready-made CSS classes and components to create responsive and attractive websites quickly.

Q10(b). What is jQuery and why was it widely used in web development?

jQuery is a JavaScript library that makes DOM manipulation, event handling, animations, and AJAX easier. It was widely used because it reduced the amount of JavaScript code needed.

Q10(c). What is React.js and what problem does it solve in frontend development?

React.js is a JavaScript library for building user interfaces using reusable components. It helps developers efficiently create and update dynamic, interactive web pages.